2802ICT Intelligent Systems

Take-Home Exercises Part 1

- No submission for the exercises is required. However, if you want the lab demonstrator help check if your answers are correct or not, please do so in the lab/workshop.
- A set of my answers will posted onto the course website in L@G.

AI and Agents

1. Based on thinking vs acting and humanly vs rationally, what are the four major views in Artificial Intelligence? Name and explain each of them. Which approach is currently taken by most AI researchers?
2. What is the Turing test? What are its contributions and problems? Which of the four major approaches in AI does it take? What is your comment on Turing test?
3. Name two most significant achievements in AI and explain why they are significant.
4. What is an agent? What do you mean by a rational agent?

Search Strategies

1. What are the four desirable properties of search algorithms? Describe what is meant by each of these four properties.
2. Give the initial state, goal test, transition (or successor function or actions) and path cost for each of the following. Choose a formulation that is precise enough to be implemented.
   (1) You have to colour a planar map using only four colours, in such a way that no two adjacent regions have the same colour.
   (2) You have three jugs, measuring 12 litres, 8 litres and 3 litres, and a water tap. You can fill the jugs up or empty them out from one to another or onto the ground. If all three jugs are full initially, you need to measure exactly one litre.
3. Breadth-first, depth-first and uniform-cost search are all considered to be *uninformed search* strategies.
   i) Explain using words and diagrams how each of these three search mechanisms operates.
   ii) Why are these algorithms considered uninformed?
   iii) Describe iterative-deepening depth-first search, in terms of its relationship with depth-first search. What are the principal advantages of iterative-deepening depth-first search over depth-first search.
4. Does each of the following statements holds or not? If yes, explain why; if not, give a counterexample:
   1) Breadth-first search is a special case of uniform-cost search.
   2) Depth-first search is a special case of best-first search.
   3) Uniform-cost search is a special case of A* search.
5. All Roads Lead to Rome
   This map shows approximate mean driving times (in hours) between pairs of cities. For each of the following graph search strategies, *work out the order in*

which states are expanded, as well as the path returned by graph search. In all cases, assume ties resolve in such a way that states with earlier alphabetical order are expanded first. The start and goal states are Warsaw and Rome, respectively. Remember that in graph search, a state is expanded only once.

(a) Depth-first search.

(b) Breadth-first search.

(c) Uniform cost search.

(d) Greedy search with the heuristic: h(Odesa) = 20 hrs; h(Budapest) = 12 hrs; h(Munich) = 3 hrs; h(Venice) = 3 hrs; h(Rome) = 0 hrs; h(Warsaw) = 30 hrs.

(e) A* search with the same heuristic.

(f) Is the heuristic admissible? Consistent? Briefly justify your answers.

(g) Which of the following is required for A* graph search to return an optimal solution?

(i) Admissible

(ii) Admissible and consistent.

(iii) It does not depend on the choice of heuristic functions, it always returns the optimal solution.

Fig. 1

6. A* search is the most well-known informed search strategy.
   i) Outline the A* search algorithm.
   ii) In general A* search is not optimal. Give a sufficient condition for the optimality of A* search.

iii) Trace the operation of A* search applied to the problem of getting to Bucharest from Lugoj using the straight-line distance heuristic. That is, show the sequence of nodes that the algorithm will consider and the f, g, and h score for each node.

iv) A robot wishes to move from square 0 ("START") to square 22 ("GOAL") in Figure 2. The following rules apply:

• The robot can only move in one of the four cardinal directions (left, right, up, down).
• The dark lines are obstacles that the robot cannot penetrate.
• The search state is defined by the square at the end of the path. Use the square number to identify that state.
• Use the Manhattan distance (ignoring obstacles) as the admissible heuristic.
• All else being equal, the A_ search is required to expand the node associated with the square with the smallest id.
• Goal test is performed on a node as part of expanding that node.
Use A* search to find a path.

Note: For each of the above A* search problems, show the following working details.

i) *draw the search tree for the final step (i.e., when a goal state is expanded),*
ii) *write down the order in which states are expanded, and*
iii) *write down the final path returned by graph search.*

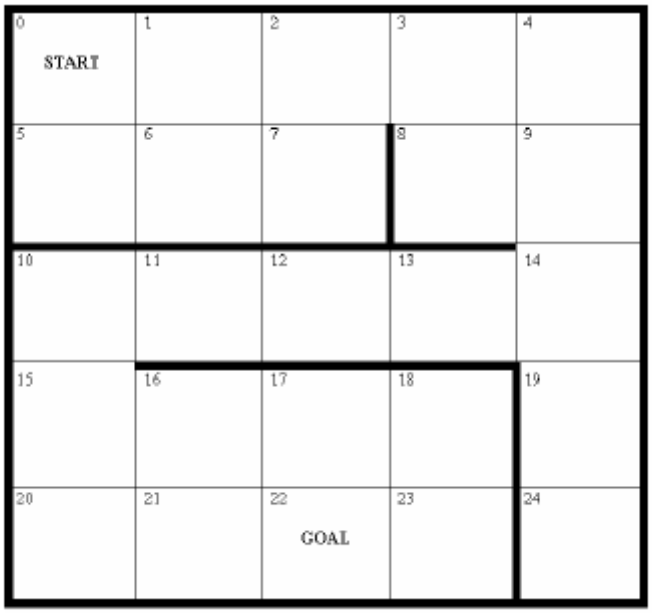

Fig. 2

7. Hill-Climbing is a well-known local search method. Using the *n*-queens problem as an example, explain how a hill-climbing local search works.